InsightEngine: An End-to-End Data Engineering Platform for Business Intelligence

Team Members :

S.NO	NAME	ROLL NUMBER
1	Somu Vaishnavi	2420030024
2	T. Dhanya Sri	2420030338
3	M. Srihitha	2420030405

Faculty Guide : Dr. N Sirisha

Overview

Abstract

InsightEngine is an end-to-end data engineering platform designed to collect, ingest, process, transform, store, and analyze data from multiple business sources to support data-driven decision-making. The platform simulates real-world data engineering workflows by generating structured and semi-structured business data using Python and ingesting it through batch and streaming mechanisms. Apache Kafka is used for data ingestion, PostgreSQL is used for operational and analytical storage, and PySpark is used for scalable data processing and transformations. Apache Airflow orchestrates and automates the complete data pipeline, including data validation, cleaning, transformation, and loading into a structured data warehouse based on dimensional modeling principles. The processed data is presented through an interactive Streamlit dashboard to provide meaningful business insights. The project demonstrates key data engineering concepts including source systems, data architecture, storage systems, ingestion strategies, batch processing, data modeling, transformations, and the complete data engineering lifecycle.

Introduction

In today's data-driven environment, organizations generate large volumes of data from multiple sources such as customer transactions, products, payments, inventory, and business operations. Raw data from these sources is often distributed, inconsistent, and difficult to use directly for decision-making. Data engineering provides the processes and technologies required to collect, ingest, store, process, transform, and organize this data into reliable and meaningful information.

InsightEngine is an end-to-end data engineering platform designed to demonstrate the complete data engineering lifecycle. The platform generates realistic business data using Python, supports batch and streaming data ingestion through Python and Apache Kafka, stores operational data in PostgreSQL, and performs data cleaning and transformation using PySpark. Apache Airflow is used to orchestrate and automate the pipeline, while the transformed data is stored in a structured data warehouse using dimensional modeling principles. Finally, an interactive Streamlit dashboard presents business insights. The project integrates key data engineering concepts including source systems, data architecture, storage, ingestion, batch processing, data modeling, transformations, and data quality.

Objectives

- Design an end-to-end data engineering pipeline for handling business data.
- Implement batch and streaming data ingestion using Python and Kafka.
- Store and manage data efficiently using PostgreSQL.
- Clean, validate, and transform data using Python and PySpark.
- Build a structured data warehouse using dimensional modeling.
- Automate the data pipeline using Apache Airflow.
- Generate meaningful business insights through an interactive Streamlit dashboard.

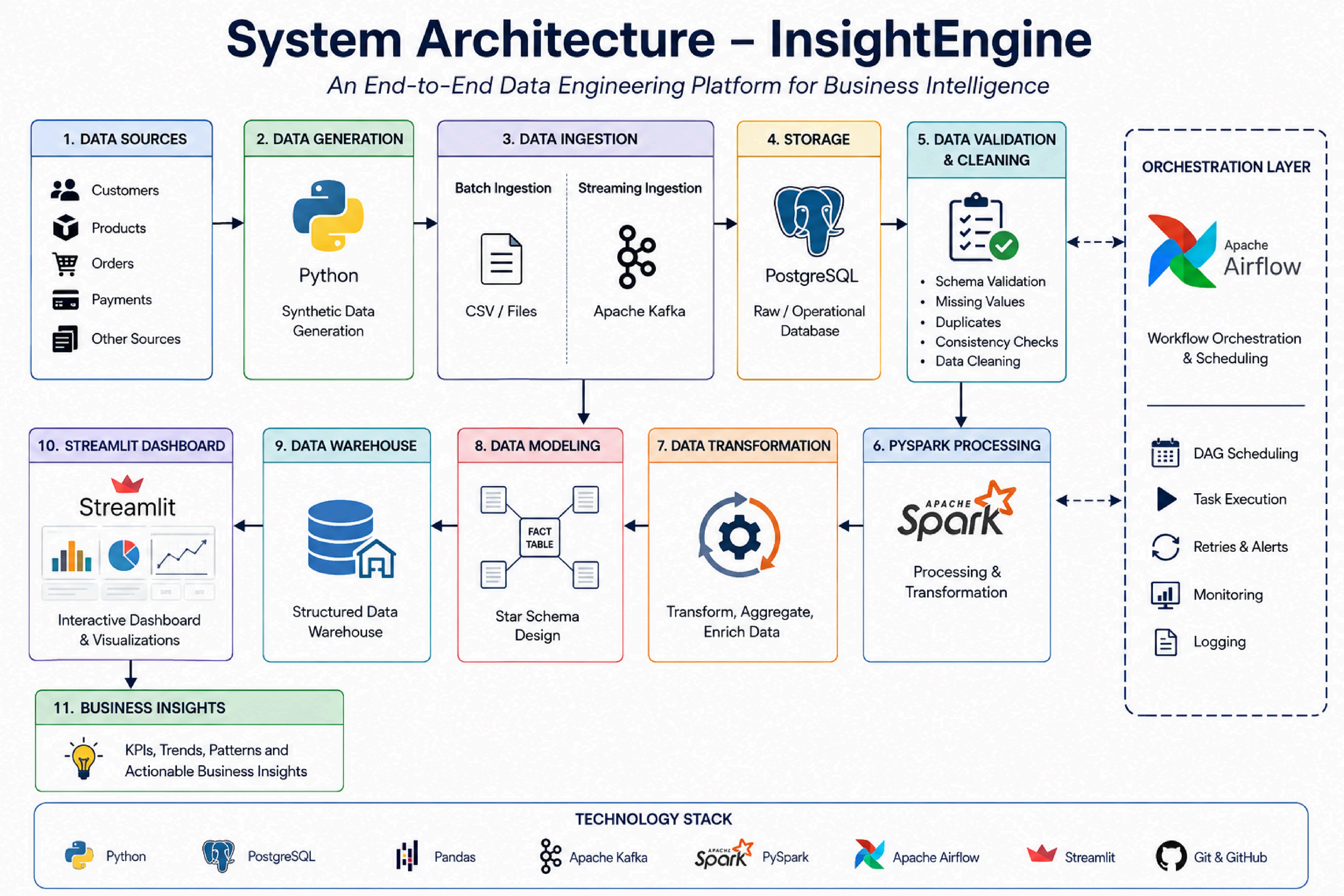
Literature Survey

S.No	Reference Link	Authors	Drawbacks	Future Work
1	DOI / PAPER	T. Akidau et al.	Balancing correctness, latency and cost is complex	Improve optimization for large-scale streaming
2	Paper / PDF	J. Kreps, N. Narkhede, J. Rao	Focuses mainly on distributed log processing	Improve scalability and system integration
3	DOI / Paper	M. Zaharia et al.	Distributed processing can require significant resources	Improve efficiency and resource utilization
4	DOI / Paper	S. Melnik et al.	Designed primarily for web-scale analytical workloads	Extend scalable interactive analytics
5	DOI / Paper	M. Gribaudo, M. Iacono, M. Kiran	Separate batch and real-time paths increase complexity	Improve resource allocation and performance

Literature Survey

S.No	Reference Link	Authors	Drawbacks	Future Work
6	IEEE / DOI	R. Hai et al.	Data-lake management and metadata are complex	Improve metadata management and integration
7	DOI / Paper	H. Foidl et al.	Data cleaning and integration cause quality issues	Improve automated data-quality mechanisms
8	DOI / Paper	J. Yasmin et al.	Workflow configuration and execution are challenging	Improve workflow tooling and usability
9	DOI / Paper	A. Simitsis et al.	ETL development is complex and resource-intensive	Improve automated ETL design and optimization
10	DOI / Paper	Mallikarjuna Rao Vasa	Focuses heavily on GenAI/agentic automation; requires complex multi-agent architecture	Extend autonomous ETL, governance and multi-domain data engineering

System Architecture



Thank You